STUDENT PERFORMANCE ANALYSIS REPORT

Executive Summary

This project analyzes student academic performance and attendance records to identify factors affecting learning outcomes. The objective was to evaluate subject performance, attendance trends, and pass/fail rates.

Dataset Overview

The dataset includes student marks in multiple subjects and attendance percentages. Analysis was performed using Python and statistical techniques.

Methodology

The following process was followed:

1. Data collection and loading
2. Data quality assessment
3. Calculation of average scores
4. Pass/fail classification
5. Attendance analysis
6. Visualization and insight generation

Key Findings

- Attendance positively impacts academic performance.
- Students with higher attendance generally achieve better scores.
- Subject-wise performance varies significantly.
- Most students achieved passing grades.
- Attendance can be used as an early performance indicator.

Visualizations Generated

- Subject-wise Average Scores
- Attendance Distribution
- Pass/Fail Ratio

Business Insights

1. Consistent attendance contributes to improved academic performance.
2. Certain subjects require additional academic support.
3. Student engagement strongly influences learning outcomes.
4. Attendance monitoring can help identify at-risk students.
5. Performance trends provide opportunities for intervention programs.

Recommendations

- Improve attendance tracking systems.
- Conduct additional support sessions for low-performing students.
- Encourage active participation in classroom activities.
- Introduce performance monitoring dashboards.
- Provide personalized learning assistance where required.

Conclusion

The analysis demonstrates a clear relationship between attendance and academic performance. Educational institutions can use these findings to improve student success rates and learning outcomes.